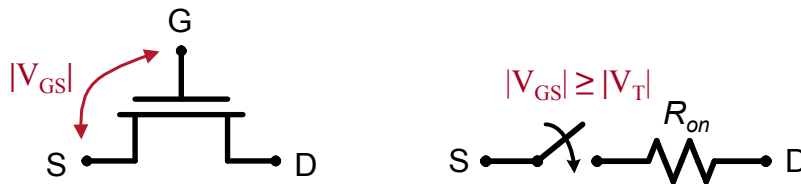


What is a Transistor?

An MOS Transistor $\longleftrightarrow$ A Switch!

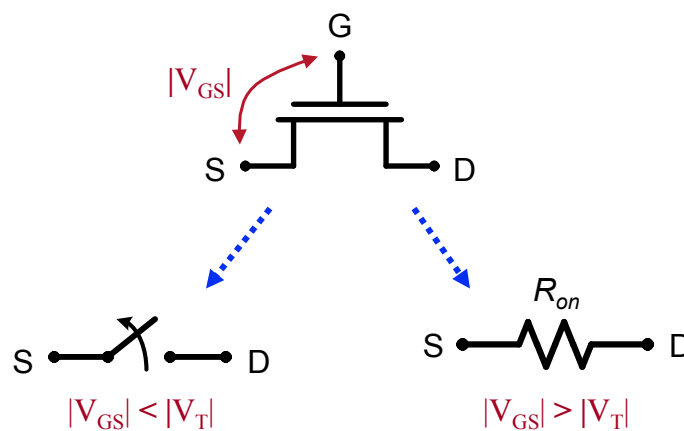


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Switch Model of MOS Transistor

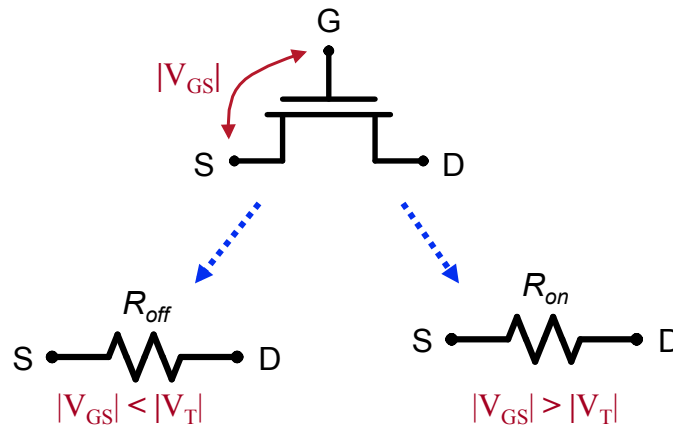


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A Modern Sub-100 nm Look ...



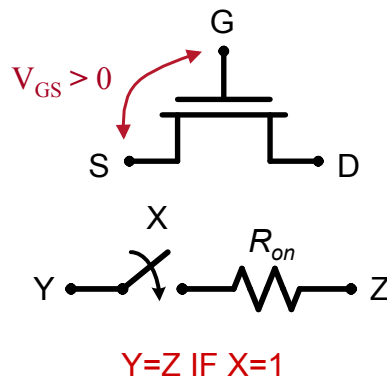
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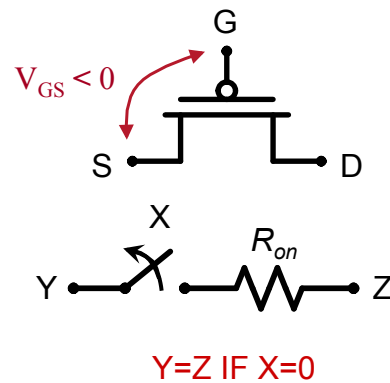
25

NMOS and PMOS

NMOS Transistor



PMOS Transistor



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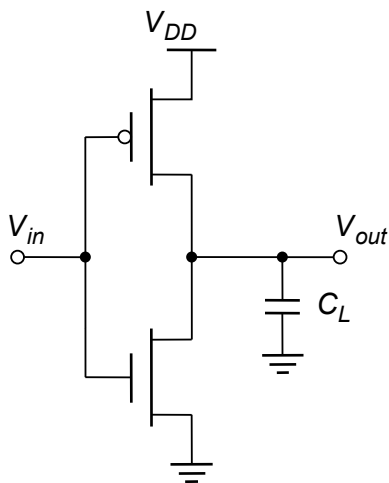
Building an Inverter with Switches

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The CMOS Inverter: A First Glance

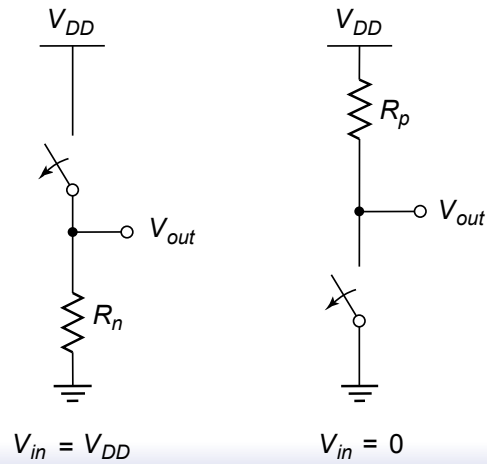


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CMOS Inverter First-Order DC Analysis



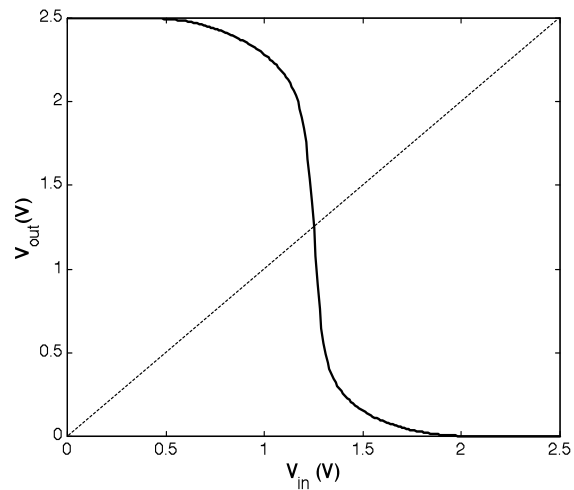
$$\begin{aligned} V_{OL} &= 0 \\ V_{OH} &= V_{DD} \\ V_M &= f(R_n, R_p) \end{aligned}$$

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Simulated Inverter VTC (Spice)



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CMOS Inverter: DC Properties

- $V_{OH} =$
- $V_{OL} =$
- $V_{IL} =$
- $V_{IH} =$
- $N_{MH} =$
- $N_{ML} =$
- $V_M =$

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CMOS Inverter: DC Properties

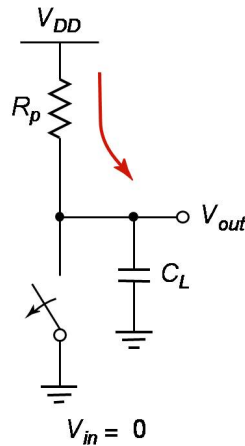
- $V_{OH} = V_{DD} = 2.5V$
- $V_{OL} = 0V$
- $V_M = 1.2V$
- $V_{IL} = 1.05V$
- $V_{IH} = 1.45V$
- $N_{MH} = 1.05V$
- $N_{ML} = 1.05V$

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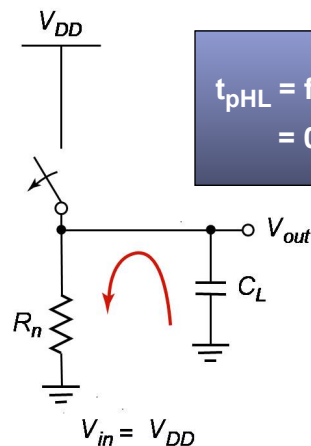
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CMOS Inverter: Transient Response



(a) Low-to-high

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(b) High-to-low

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$$t_{pHL} = f(R_{on}C_L) \\ = 0.69 R_n C_L$$

CMOS Properties

- ❑ Full rail-to-rail swing
- ❑ Symmetrical VTC
- ❑ Propagation delay function of load capacitance and resistance of transistors
- ❑ No static power dissipation
- ❑ Direct path current during switching

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Impact on Reliability?

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Impact on Performance?

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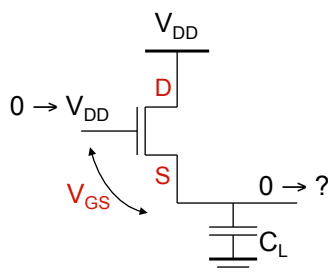
Impact on Power/Energy

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A Side Note: How Good a Switch is the MOS Transistor?

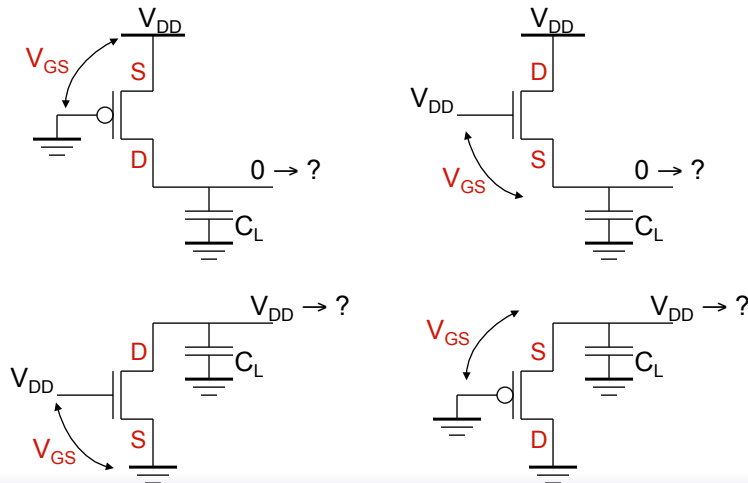


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How Good a Switch is an MOS?

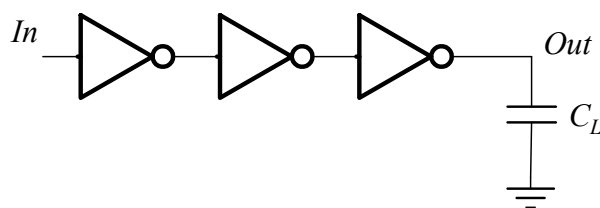


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The Next Question: Inverter Chain



- For some given C_L :
 - How many stages are needed to minimize delay?
 - How to size the inverters?
- Anyone want to guess the solution?

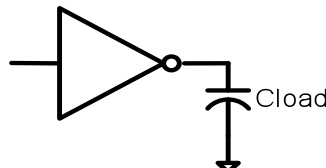
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Careful about Optimization Problems

- ❑ Get fastest delay if build one **very** big inverter
 - So big that delay is set only by self-loading



- ❑ Likely not the problem you're interested in
 - Someone has to drive this inverter...

Engineering Optimization Problems in General

- ❑ Need to have a set of constraints
- ❑ Constraints key to:
 - Making the result useful
 - Making the problem have a 'clean' solution
- ❑ For sizing problem:
 - Need to constrain size of first inverter

Delay Optimization Problem #1

- You are given:
 - A fixed number of inverters
 - The size of the first inverter
 - The size of the load that needs to be driven
- Your goal:
 - Minimize the delay of the inverter chain
- Need model for inverter delay vs. size

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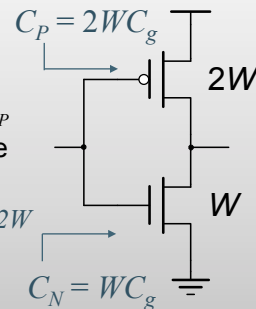
Inverter Delay

Delay: $t_{pHL} = (\ln 2) R_N C_L$ $t_{pLH} = (\ln 2) R_P C_L$

- Assume we want equal rise/fall delays

$$t_{pHL} = t_{pLH}$$

- Need approximately equal resistances, $R_N = R_P$
- PMOS approximately 2 times larger resistance for same size;
- Must make PMOS 2 times wider, $W_P = 2W_N = 2W$
- $t_p = (\ln 2) (R_{inv}/W) C_L$ with R_{inv} resistance of minimum size NMOS



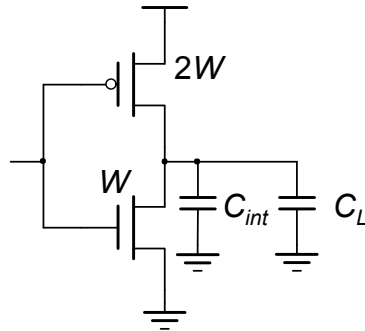
Loading on the previous stage: $C_{in} = WC_{ginv} = W(3C_G)$

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Inverter Delay Model



$$R = R_{inv}/W$$

$$C_{int} = W(3C_d) = WC_{dinv}$$

Replace $\ln(2)$ with k (a constant):

$$\text{Delay} = kR(C_{int} + C_L)$$

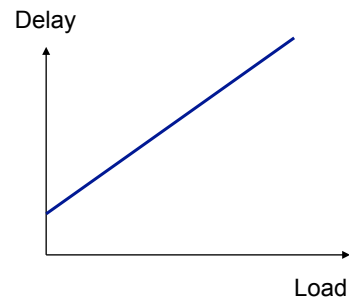
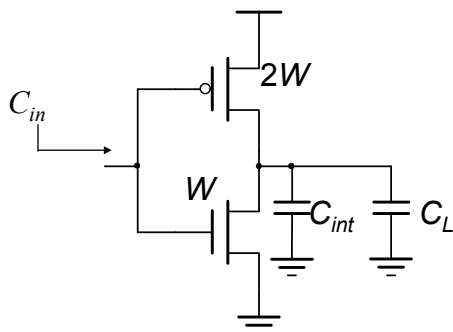
$$\text{Delay} = k(R_{min}/W)(WC_{dinv} + C_L)$$

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Inverter with Load



$$\begin{aligned} \text{Delay} &= kR C_{in}(C_{int}/C_{in} + C_L/C_{in}) \\ &= kR_{min} C_{ginv} [C_{dinv}/C_{ginv} + C_L/(WC_{ginv})] \\ &= \text{Delay (Internal)} + \text{Delay (Load)} \end{aligned}$$

$$C_{dinv}/C_{ginv} = \gamma = \text{Constant independent of size}$$

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Delay Formula

$$\text{Delay} \sim R_W (C_{int} + C_L)$$

$$t_p = kR_W C_{in} (C_{int}/C_{in} + C_L / C_{in}) = t_{inv} (\gamma + f)$$

$$C_{int} = \gamma C_{in} (\gamma \approx 1 \text{ for CMOS inverter})$$

$$f = C_L / C_{in} - \text{electrical fanout}$$

$$t_{inv} = kR_{min} C_{ginv}$$

t_{inv} is independent of sizing of the gate!!!